

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	MAS-ITEC (Coursera)	
Program Name	Master of Information Technology Management	
Assessment Year	2024-2025	
Program Assessment Coordinator	Marwan Omar	
Year of Current Assessment Plan	2025	
No Assessment This Year per Plan	☐	
LO(s) Assessed	LO #2	Manage and deploy secure cloud-based information resources in an enterprise setting
	LO #3	Pursue a diverse range of careers in cloud computing
Presented at program authority meeting date: <i>Please attach minutes</i>		

LO #2 – Manage and deploy secure cloud-based information resources in an enterprise setting

The selected artifact was the **Photo Sharing Application Project**. This project required students to implement a secure, cloud-based photo sharing platform using AWS services such as **Cognito, S3, CloudFront, Lambda, API Gateway, and DynamoDB**. Students were assessed on their ability to configure authentication, enable encryption, enforce access policies, and design secure storage/distribution mechanisms.

LO #3 – Pursue a diverse range of careers in cloud computing

The artifact chosen was also the **Photo Sharing Application Project** with the addition of a **professional report component**. Students were required to produce deliverables demonstrating their ability to communicate cloud security design decisions in a professional manner, including **architecture diagrams, threat models (STRIDE and OWASP Top 10), and mitigation strategies**. These skills align directly with career preparation for roles in cloud security engineering, cloud architecture, and compliance analysis.

Rubrics used:

- **LO2:** Technical correctness, completeness of AWS implementation, use of secure configurations, ability to integrate multiple cloud services.
 - **LO3:** Clarity and professionalism of documentation, application of industry frameworks, ability to contextualize technical work within career-relevant roles.
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Results

Assessment showed strong overall performance:

- **LO2:** Students demonstrated high proficiency in deploying secure cloud systems, particularly in configuring Cognito for identity management, encrypting S3 storage, applying DynamoDB policies, and securing CloudFront delivery

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- **LO3:** Students produced professional reports including **architecture and data flow diagrams**

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, threat modeling using STRIDE/OWASP

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, and recommendations for security best practices. However, depth and quality of professional writing varied across submissions.

Artifact Samples

Representative student submissions were reviewed:

- **Artifact 1 ():** Detailed step-by-step AWS Cognito and S3 configuration with screenshots; strong technical implementation
- **Artifact 2 ():** Comprehensive report including **architecture diagrams, STRIDE threat modeling, OWASP-based mitigations, and WAF configuration**
- **Artifact 3 ():** Solid technical work with Cognito, S3, and DynamoDB; professional reporting was less comprehensive
- **Artifact 4 ():** Demonstrated core technical deployments but with less detail in professional documentation

Data Table

Distribution of student performance (n=4 sample set):

Learning Outcome	Does Not Meet	Meets	Proficient
LO2 (Secure cloud deployment)	0	1	3
LO3 (Career readiness in cloud)	0	2	2

Summary: All students met or exceeded expectations. The majority reached proficiency in cloud deployment (LO2), while results for LO3 were more evenly split between “Meets” and “Proficient,” reflecting variability in professional writing and communication.

Description of Findings

The program authority reviewed the assessment results and discussed strengths and areas for improvement. The discussion emphasized that **students are successfully learning to manage and deploy secure cloud environments using AWS**. The artifacts demonstrated strong technical integration of identity management, encryption, and cloud storage.

However, it was also noted that **professional communication skills varied across students**. Some produced excellent consulting-style reports with clear architecture diagrams and industry-aligned threat modeling, while others focused more heavily on technical implementation with less emphasis on professional presentation.

Areas of concern: No major concerns were identified, but the faculty agreed that improving consistency in professional documentation and report writing would enhance career readiness (LO3).

This Cycle's CI Plan

Based on the assessment of ITMS 564 artifacts, students demonstrated strong ability to deploy secure cloud applications (AWS Cognito, S3, DynamoDB, CloudFront) and integrate identity, encryption, and policy controls. Graduate-level work included architectural diagrams, threat modeling (STRIDE/OWASP), and compliance considerations. While proficiency was high overall, the following areas for continuous improvement were identified:

1. **Expand Threat Modeling and Compliance Integration:**
Students successfully applied STRIDE and OWASP but often limited coverage to technical threats. Beginning in **AY26**, we will introduce labs requiring integration of compliance frameworks (e.g., FedRAMP, HIPAA, GDPR) into threat modeling exercises.
2. **Enhance Career Preparation Through Applied Projects:**
Artifacts showed strong technical builds but uneven professional reporting. To strengthen career readiness (LO #3), we will require deliverables in the format of **professional consulting reports** and include a **mock client presentation** exercise.
3. **Introduce Cross-Cloud Exposure:**
Projects were focused primarily on AWS. Starting in **AY26**, an optional comparative module on Azure or GCP will be piloted to broaden student skills and reflect industry demand for multi-cloud proficiency.

Timeline for Implementation:

- **AY26:** Pilot compliance-integrated labs and professional reporting rubric.
- **AY27:** Add optional cross-cloud module and evaluate outcomes.

Responsible Parties:

- **Program Coordinator (Dr. Marwan Omar)** will oversee implementation.
- **Course Faculty teaching ITMS 564** will design the enhanced labs and rubrics.
- **Assessment Committee** will evaluate results and adjust in the following cycle.